



Design of Algorithms

CMSC 18 | 1S SY2026-27



Zazheska Lorenzo

UP MIN - CSM - DMPCS



Topics for today's session

- Introduction to Algorithm Development
- Pseudocode
- Flowchart



Programming Mindset

- Programming is **not just about coding**.
- Programming is solving problems systematically.
- Code is to a programmer as a paintbrush is to a painter.



— Example Problem



You are sitting in your living room, watching a show you like.

Eventually, you get hungry and are craving a simple ham & cheese sandwich.

But, you still want to continue watching TV afterwards.

List a set of instructions that would satisfy you in this given scenario.

PROGRAMMING MINDSET: BE SPECIFIC WITH YOUR INSTRUCTIONS



COMPUTERS CANNOT THINK FOR THEMSELVES!



Algorithms

- a **precise, ordered** set of steps used to accomplish a task.
- a good algorithm has the following characteristics:
 - finite
 - unambiguous
 - efficient



Algorithm $\neq$ Program

- Although algorithms are present in a program, they are not the same/interchangable with one another.
- Algorithms = how to solve problem
- Program = solution expressed in programming language



Software Development Process

REQUIREMENTS SPECIFICATION

State the problem. Gain a clear understanding of what is required for its solution

ANALYSIS

Identify the problem's inputs, desired outputs, and any additional requirements or constraints on the solution.

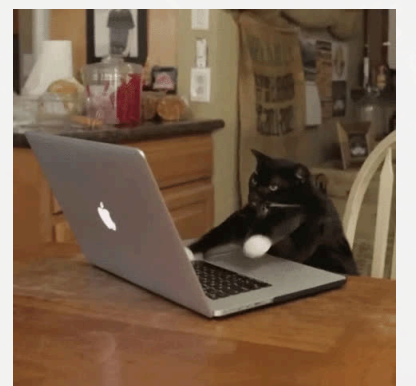
INPUT



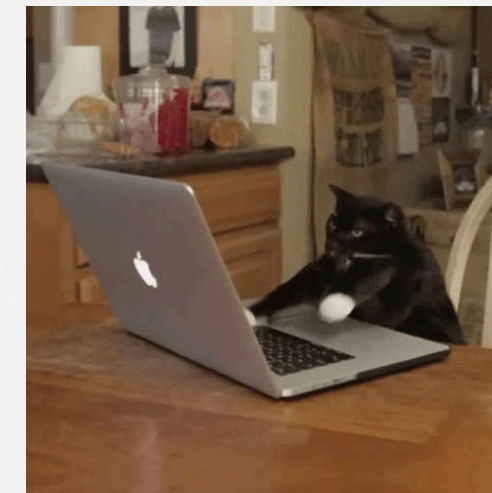
PROCESS



OUTPUT



Development Process



DESIGN

Design and develop set of steps (algorithm/s) to solve the problem

IMPLEMENTATION

Implement the algorithm as a program.

VERIFICATION AND TESTING

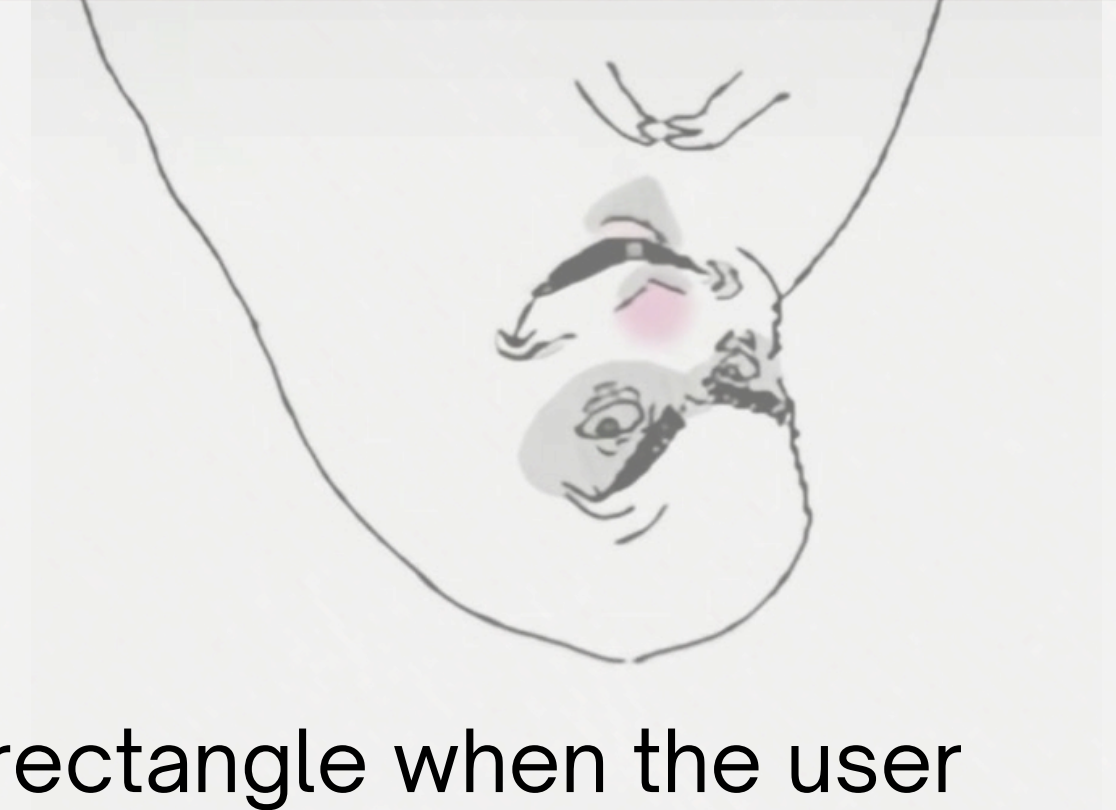
Test the completed program and verify that it works as expected.

Algorithm Representation

- Pseudocode
 - Algorithms written in a structured English method
 - Steps are outlined
- Flowchart



Pseudocode Example



- **Problem:** Design a program that outputs the area of a rectangle when the user enters its length and width.

- **Analysis:**

- Inputs: length, width
- Output: area of rectangle
- Process: $A = L \times W$

- **Pseudocode**

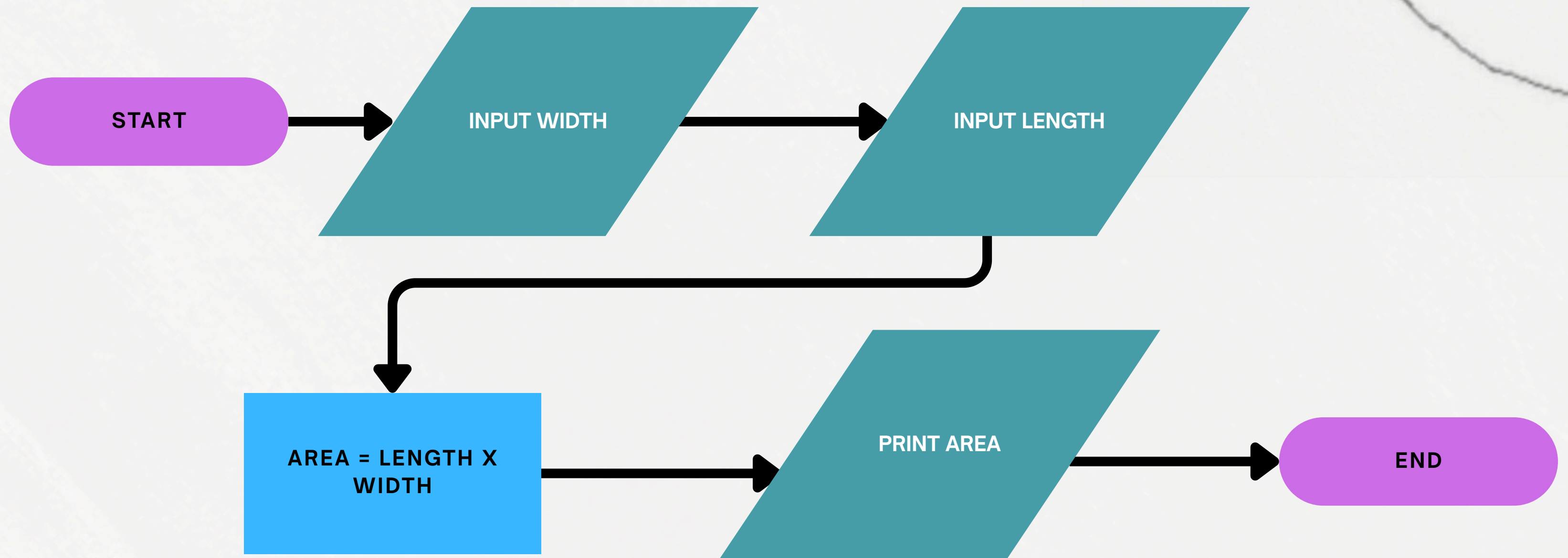
1. Get the length through user input
2. Get the width through user input
3. Multiply the length with the width
4. Print the product

Algorithm Representation

- Pseudocode
- Flowchart
 - Structured picture map showing the steps of an algorithm
 - Uses shapes and figures
 - Easy to identify the input, output, processes and flow of the algorithm



Flowchart Example



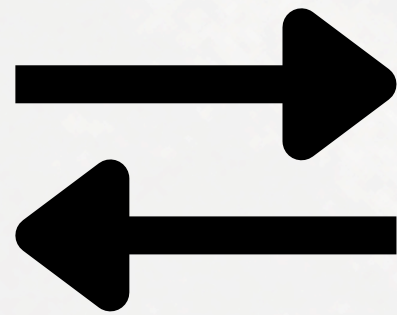
Flowchart Components



START/END/STOP/EXIT



PROCESSING



FLOWLINES



INPUT/OUTPUT



DECISION



QUESTIONS? CONCERNS?

(VIOLENT REACTIONS?)



The End

THANK YOU FOR LISTENING

Zazheska Lorenzo

UP MIN - CSM - DMPCS

